

# AtmosCare: Infrastructure Resiliency Tracker

## Database Systems – Group Project Report

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## 1. Introduction & Project Goal

### Project Overview

AtmosCare is a database-driven environmental health monitoring system that integrates real-time weather data, air quality metrics, community demographics, and vulnerability indicators. The application enables users to analyze environmental conditions and assess public health risks in specific regions.

### Goal of the Project

The primary objective of the project is to demonstrate how structured data storage, relational modeling, and analytical SQL queries can collectively help underserved populations overcome informational barriers regarding environmental conditions. The project showcases how databases can transform raw weather and demographic data into actionable insights.

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## 2. Database Design

### 2.1 Schema Overview

The AtmosCare database consists of four primary tables:

1. **LOCATION** – Defines geographic regions and vulnerability attributes.
2. **WEATHER\_RECORD** – Stores environmental observations across time.
3. **HEALTH\_RISK** – Defines health risks and relevant environmental thresholds.

4. **RISK\_ALERT** – Connects weather records to health risks.

These tables support both real-time ingestion and historical trend analysis.

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## 2.2 Table Structures

### LOCATION

| Column              | Type    | Description                   |
|---------------------|---------|-------------------------------|
| zip_code (PK)       | VARCHAR | Unique location identifier    |
| location_name       | TEXT    | Community name                |
| population          | INT     | Estimated population          |
| vulnerability_index | FLOAT   | Risk level for that community |

### WEATHER\_RECORD

| Column            | Type      | Description          |
|-------------------|-----------|----------------------|
| record_id (PK)    | SERIAL    | Unique weather entry |
| zip_code (FK)     | VARCHAR   | References LOCATION  |
| timestamp         | TIMESTAMP | Time of observation  |
| temperature       | FLOAT     | Temperature reading  |
| humidity          | FLOAT     | Percent humidity     |
| air_quality_index | INT       | EPA AQI category     |

### HEALTH\_RISK

| Column         | Type    | Description                          |
|----------------|---------|--------------------------------------|
| risk_id (PK)   | SERIAL  | Unique health risk identifier        |
| risk_name      | VARCHAR | Example: “Asthma”, “Heat Stroke”     |
| threshold_type | VARCHAR | Logical rule applied to weather data |

### RISK\_ALERT

| Column         | Type    | Description                      |
|----------------|---------|----------------------------------|
| alert_id (PK)  | SERIAL  | Unique alert                     |
| zip_code (FK)  | VARCHAR | Affected location                |
| record_id (FK) | INT     | Weather record meeting threshold |
| risk_id (FK)   | INT     | Triggered health risk            |
| is_urgent      | BOOLEAN | Whether conditions are severe    |

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### 2.3 Entity-Relationship Model Summary

- LOCATION (1) = WEATHER\_RECORD ( $\infty$ )
- WEATHER\_RECORD (1) = RISK\_ALERT ( $\infty$ )
- HEALTH\_RISK (1) = RISK\_ALERT ( $\infty$ )

This structure allows linking real-time environment data to vulnerability profiles and health conditions.

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### 2.4 Assumptions

- ZIP codes uniquely identify geographic locations.
  - Vulnerability indices are precomputed and static.
  - WeatherAPI provides reliable real-time environmental metrics.
  - Health thresholds apply uniformly to all communities.
  - Time granularity of weather data does not need sub-minute precision.
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## 3. System Implementation

### 3.1 Backend Implementation

- Implemented using **FastAPI**
- PostgreSQL database for persistent data storage
- Asynchronous ingestion of WeatherAPI data
- Routes implemented for CRUD operations and advanced analytical queries
- Parameterized SQL queries ensure security and scalability

### 3.2 Frontend Implementation

- Built with **React + Vite + TypeScript**
- Component-based interface for Locations, Weather, and Analytics
- API communication implemented via reusable `http()` wrapper
- Designed to visualize database-driven results clearly and interactively

### 3.3 Analytical Queries Implemented

1. **High-Risk Population Exposure Query**  
Identifies ZIP codes where AQI exceeds thresholds *and* vulnerability is high.
  2. **Historical Heat Outlier Query**  
Finds ZIP codes with the most days exceeding the 95th percentile temperature.
  3. **Proactive Health Alert Query**  
Maps real-time conditions to specific health risks using threshold rules.
  4. **Resource Allocation Query**  
Calculates urgent alert percentages for high-risk vs. low-risk communities.
  5. **Comparative Weather Query**  
Compares averages for two ZIP codes over a rolling 7-day period.
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## 4. Breaking Barriers: Results & Impact

### 4.1 Insights Generated

The dataset enables:

- Identification of **environmentally vulnerable communities**
- Detection of **extreme heat trends** across multiple regions
- Real-time mapping of environmental hazards to **health risks**
- Clear understanding of **alert frequency and distribution**
- Comparisons between communities to expose inequities

### 4.2 How the Dataset Breaks Barriers

The system breaks informational barriers by:

- Making climate-health risks **visible and understandable**
- Providing accessible real-time environmental insights
- Connecting environmental data to **public health outcomes**
- Supporting equitable resource allocation
- Empowering vulnerable populations through early warnings

These insights transform raw environmental data into actionable intelligence, supporting community resilience and health preparedness.

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## 5. Conclusion

AtmosCare showcases how relational databases can serve as powerful tools in public health analytics. Through careful schema design, real-time data ingestion, and strategic SQL queries, the system reveals environmental hazards and population vulnerabilities that would otherwise remain hidden.

By organizing and analyzing environmental data, AtmosCare effectively breaks barriers in public health awareness, providing timely insights that can help protect at-risk populations.

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